

|   |                                                                                                                                                          |                              |     |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-----|
| 3 | (a) point where the weight of an object / gravitational force may be considered to act                                                                   | M1<br>A1                     | [2] |
|   | (b) product of the force and the <u>perpendicular</u> distance (to the pivot)                                                                            | B1                           | [1] |
|   | (c) (i) 1. sum / net / resultant force is zero                                                                                                           | B1                           |     |
|   | 2. net / resultant moment is zero<br>sum of clockwise moments = sum of anticlockwise moments                                                             | B1                           | [2] |
|   | (ii) $W \times 0.2 = 80 \times 0.5 + 70 \times 1.3$<br>$= 40 + 91$<br>$W = 655 \text{ N}$<br>(allow 2/3 for one error in distance but 0/3 if two errors) | C1<br>C1<br>A1               | [3] |
|   | (iii) move pivot to left<br>gives greater clockwise moment / smaller anticlockwise moment<br>or<br>move W to right<br>gives smaller anticlockwise moment | (M1)<br>(A1)<br>(M1)<br>(A1) | [2] |
| 4 | (a) (i) stress is force / area                                                                                                                           | B1                           | [1] |